Fiscal Year:	FY 2022	**Task Last Updated:**	FY 02/27/2023
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	01/27/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/27/23 per NSSC information (Ed., 2/7/23) NOTE: End date changed to 8/31/2022 per NSSC information (Ed., 2/6/22) NOTE: Project extended to 8/31/2021 with new grant 80NSSC20K1176, which has period of performance of 9/1/2020-8/31/2021 (Ed., 11/5/20) NOTE: Extended to 3/01/2021 per NSSC information (Ed., 2/27/2020) NOTE: Extended to 2/29/2020 per NSSC information (Ed., 10/16/19) NOTE: Extended to 10/31/2019 per NSSC information (Ed., 5/10/19) NOTE: Extended to 4/30/2019 per NSSC information (Ed., 4/11/18) NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lin, Hua Ph.D. (SRI International) Shaler, Thomas Ph.D. (SRI International)		

Grant/Contract No.:	80NSSC20K1176 ; NNX14AE69G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In normal biological systems, protein stability and system homeostasis are maintained through a tightly coupled post-translational modification process that is catalyzed by the ubiquitin-proteasome system for protein turnover. This process is necessary for modifying the correct protein in the correct cellular location at the correct time to control important processes, e.g., cell-cycle regulation, the maintenance of genomic stability, removal of damaged or misfolded proteins, autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights. In addition to studying early changes in protein post-translational modification after particle radiation, we extended our studies to evaluate persistent changes in pUb and specific protein phosphorylation in archival brain and heart tissues that were harvested 16 months after proton, titanium, iron and 6-beam GCRsim exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	Space particle radiation may cause significant damage to biomolecules in biological systems resulting in the formation of non-degradable aggregates or altered protein conformational folding. The presence of abundant aberrant proteins, especially in slowly regenerating tissues, can be concerning in long-term missions in space. The objective of this augmentation project is to apply the molecular techniques that were used in our previous work to evaluate persistent particle-radiation-induced changes in levels of aberrant proteins in an expanded list of the brain and heart tissues to identify persistent post-translational protein modifications in key proteins known to be involved in degenerative changes in these tissues more than a year after radiation exposure. We hypothesize that persistent oxidative stress due to the initial irradiation event results in the accumulation of damages resulting in alterations in the level of polyubiquitin (pUb) chain-modified proteins. The identification of the ubiquitin-modified proteins and the exact sites at which the target proteins are modified is predictive of an increased risk of specific adverse health consequences. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins in neuronal cells as well as the supportive cells that constitute the brain tissue microenvironment. We, therefore, expect that dysfunction of the cellular ubiquitin-related pathways leading to changes in the levels of aberrant proteins will also be dose- and linear energy transfer (LET)-dependent. Furthermore, such alterations are likely to precipitate changes in the native protein equilibrium, resulting in accelerated age-associated tissue pathology. Using the methodology that we have developed for in-depth analysis of ubiquitin-related pathways and networks employing liquid-chromatography coupled to a hybrid, high-resolution linear ion-trap/Orbitrap tandem mass spectrometry (LC-MS/MS) system, we have published on pUb changes in human neuronal cells and regionally-specific rodent neuronal cells after particle radiation exposure to understand the impact of radiation on the regulation of protein homeostasis in cells critical to the function of the central nervous system (CNS). We have also analyzed brain samples from rodents that were irradiated with particle beams with different beam qualities. To broaden our knowledge, we are leveraging this experience in the 'acute' response to examine 'delayed and persistent' dysfunction in post-translational modification in rodent brain and heart tissues that were harvested ~1.5 yrs post-irradiation. The results of our project will lead to a better understanding of tissue susceptibility to particle radiation exposure and the mechanistic pathways associated with the dynamic maintenance of normal protein structures and homeostasis that is relevant to tissue degeneration.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Shaler TA, Lin H, Chang PY. "Long-term effects of particle radiation attenuate the regulation of protein homeostasis and steady state levels of functional protein modification in the brain and heart." NASA Human Research Program (HRP) Investigator’s Workshop, Virtual meeting. Abstract. NASA Human Research Program (HRP) Investigator’s Workshop, Virtual meeting. , Feb-2022
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Bakke J, Lin H. "Particle radiation attenuates the regulation of protein homeostasis in the brain." NASA Human Research Program (HRP) Investigator’s Workshop, Virtual meeting. Abstracts. 2021 NASA Human Research Program (HRP) Investigator’s Workshop, Virtual meeting. , Feb-2021
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Bakke J, Lin H. "Proteomic analysis of neuronal cells and brain tissues after particle radiation." NASA Human Research Program (HRP) Investigator's Workshop (IWS), Galveston, Texas Jan 27-30, 2020. NASA Human Research Program (HRP) Investigator's Workshop (IWS), Galveston, Texas Jan 27-30, 2020. , Jan-2020

Articles in Other Journals or Periodicals	Braby L, Raber J, Chang PY, Dinges D, Goodhead D, Herr D, Hopewell J, Huff J, Krull K, Linnehan R, MacVittie T, O'Banion MK, Qin M, Root J, Rose S, Winsauer P. "Radiation Exposure in Space and the Potential for Central Nervous System Effects: Phase 2. " NCRP Report # 183: Scientific Committee SC1-24P2 panel. Nov 2019. , Nov-2019
Articles in Peer-reviewed Journals	Shaler T, Bakke J, Grover A, Lin H, Chang P. "Particle radiation induced dysregulation of ubiquitin proteasome system in neuronal stem cells." Life Sci Space Res (Amst). 2020 May;25:9-17. https://doi.org/10.1016/j.lssr.2020.02.003 ; PMID: 32414496 , May-2020
Articles in Peer-reviewed Journals	Madrid PB and Chang PY. "Accelerating space radiation countermeasure development through drug repurposing." Life Sci Space Res (Amst). 2022 Nov;35:30-35. https://doi.org/10.1016/j.lssr.2022.07.002 ; PubMed PMID: 36336366 , Nov-2022
Articles in Peer-reviewed Journals	Huff JL, Poignant F, Rahmanian S, Khan N, Blakely EA, Britten RA, Chang P, Fornace AJ, Hada M, Kronenberg A, Norman RB, Patel ZA, Shay JW, Weil MM, Simonsen LC, Slaba TC. "Galactic cosmic ray simulation at the NASA space radiation laboratory – Progress, challenges and recommendations on mixed-field effects." Life Sci Space Res (Amst). 2023 Feb;36:90-104. https://doi.org/10.1016/j.lssr.2022.09.001 ; PubMed PMID: 36682835 , Feb-2023